

Software Engineering (UE24CS341A)

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Project: PassPulse

Team 2

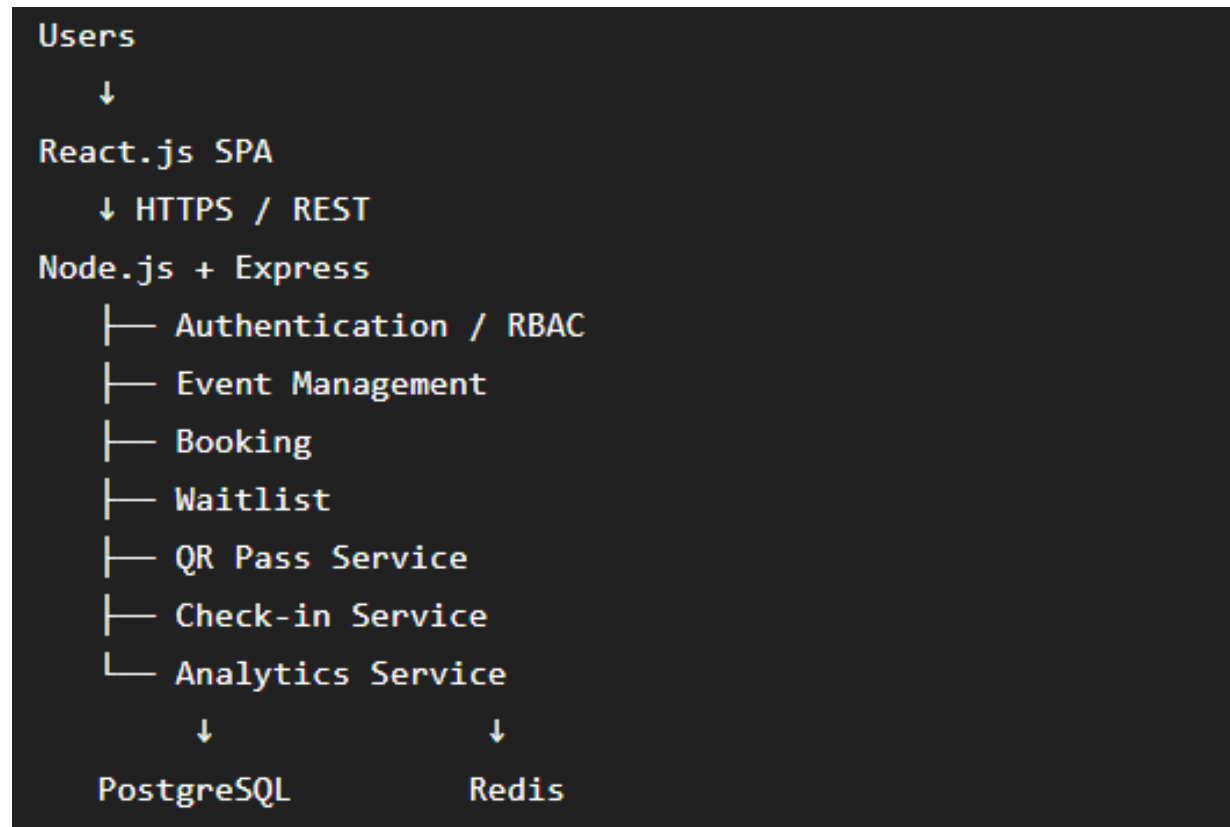
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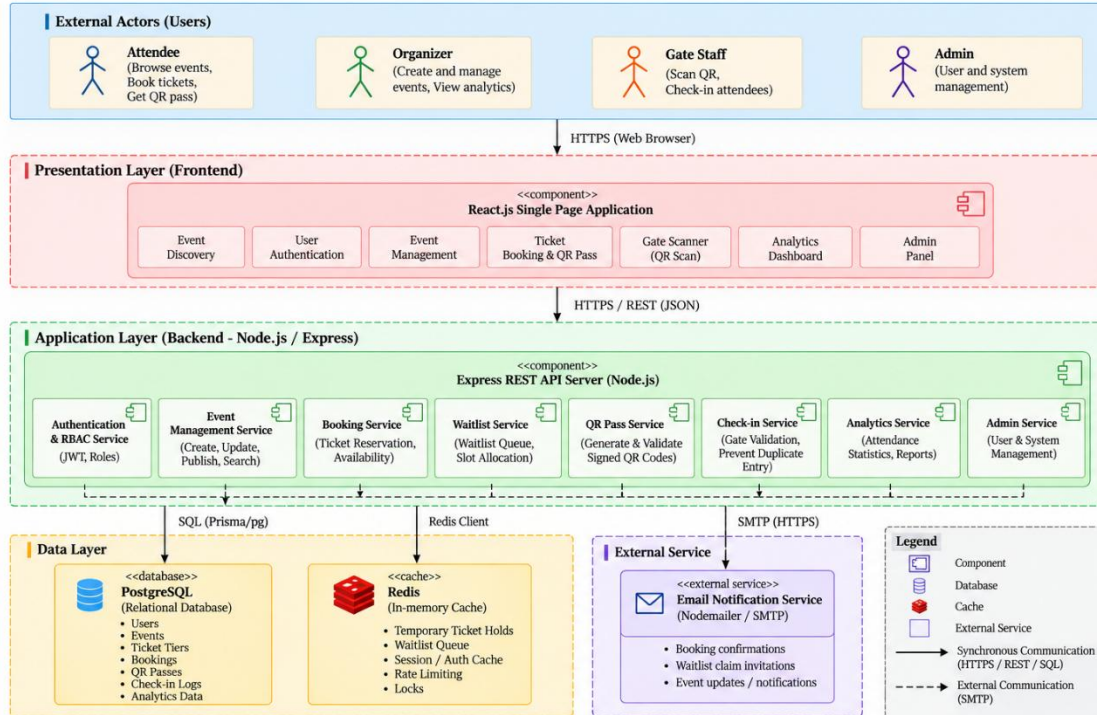
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Component Diagram:



Architectural Pattern & Justification

PassPulse – Component Diagram
(Layered Architecture)



Component Descriptions & Requirement Traceability

Component	Responsibility
React.js SPA	Provides the user-facing interface for attendees, organizers, gate staff and administrators.
Authentication & RBAC Service	Handles user authentication and role-based authorization.
Event Management Service	Handles event creation, updating, scheduling, publishing and event discovery.
Booking Service	Manages ticket reservation, availability and booking confirmation.
Waitlist Service	Manages FIFO waitlists and slot reallocation after cancellations.
QR Pass Service	Generates and validates cryptographically signed QR passes for confirmed tickets.
Check-in Service	Validates scanned QR passes, records attendance and prevents duplicate entry.
Analytics Service	Processes attendance information and provides statistics and reports.
Admin Service	Provides administrative user and event management functions.
PostgreSQL	Stores persistent user, event, ticket, booking, QR and check-in data.
Redis	Supports temporary ticket holds, locks and waitlist-related operations.
Email Notification Service	Sends booking confirmations, waitlist invitations and event notifications.

4.1 Requirement Traceability

Component	Related Requirements
Authentication & RBAC Service	FR-001, FR-002, FR-003
Event Management Service	FR-004, FR-005, FR-006
Booking Service	FR-007, FR-008, FR-009, FR-010
QR Pass Service	FR-011, FR-012
Check-in Service	FR-013, FR-014
Waitlist Service	FR-015, FR-016
Analytics Service	FR-017, FR-018
Admin Service	FR-019
Overall architecture	NFR-001, NFR-002, NFR-003

Traceability Statement:

Each major architectural component is mapped to one or more functional or non-functional requirements defined in the PassPulse SRS. This ensures that architectural decisions are directly related to specified system behavior and quality requirements.

Security Architecture:

JWT Authentication



RBAC Authorization



TLS-secured communication



Input validation



HMAC-signed QR validation



Audit logging